



EE 505: Materials for Semiconductor Industry

Course Code	EE 505
Course Name	Materials for Semiconductor Industry
Approval	8th Senate (Oct 2015)

Course Outline

Semiconductor fundamentals, band structure, indirect and direct band gap, optical properties, carrier statistics, semiconductor material purification and crystal growth, epitaxy, CVD and MBE, P-N Junction, Schottky and Mas device structures, specific material requirements, Doping by implantation and diffusion, dielectric and insulators, ohmic and barrier contacts, band edge behaviour, empirical rule, alloy design.